

Tsunami forecasting in the Makran subduction zone using machine learning surrogates

by

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Abstract

Physics based simulations for probabilistic tsunami hazard assessment are very computationally demanding and this severely limits their applicability in real time forecasting. There are many different methods that have been used to get around this issue and one of the most common is using ML surrogates for various parts of the simulation. In this project we propose to do just this for the Makran subduction zone (MSZ), we replace a Boussinesq solver with an ML surrogate using initial surface elevation snapshots, bathymetry, source earthquake slip distribution type and station coordinates as inputs. We focus on predicting the maximum wave height and the first wave arrival time for 317 stations and we train a shared ResNet style CNN encoder with an MLP decoder. We also include aleatoric uncertainty quantification via probabilistic heads. These steps deliver a substantial speedup enabling near real time crisis forecasting and the results are encouraging, especially when it comes to the probabilistic metrics. The point prediction results, however, fall short of the best in the literature. We point to additional work needed to match these while still providing a more extensive coverage of the MSZ coast and dealing with a wider variety of tsunami scenarios.

Introduction

Probabilistic tsunami hazard assessment generally involves defining a multitude of credible scenarios (preferably accounting for aleatory and epistemic uncertainty) then simulating the generation, propagation and occasionally the inundation using numerical models and finally aggregating the results often into hazard curves or exceedance maps. This method circumvents the issue of insufficient historical data (Salah et al., 2021) and allows for a comprehensive exploration of different sources of uncertainty (Ramalingam et al., 2025) which makes it extremely valuable when it comes to making risk informed decisions. However, this comes with a large computational cost and this can be a significant drawback in some cases, especially when we need near real time predictions.

A common way to get around this issue is the use of ML surrogates for the entire simulation or at least the most expensive parts of it. Once trained, these can help generate predictions in as little as milliseconds as opposed to the minutes or even hours it would take otherwise. There are lots of different ML models that have already been successfully used in this context, CNNs have shown good performance in early warning systems (Makinoshima et al., 2021) and the same goes for MLPs (Rodríguez et al., 2022) and this with inputs going from short tsunami observations to just the earthquake source parameters. There is a table in Ramalingam et al. (2025) with additional examples.

Here we propose to do something similar, we focus on the Makran Subduction Zone and we build on previous work. The dataset we use has already been generated according to the methodology defined in Salah et al. (2021). The tsunami generation was done using the Okada model and the Kajiura filter whereas the propagation was done using FUNWAVE-TVD, a Boussinesq solver. The inundation stage was however skipped. Part of our dataset has also already been used in a previous similar study (Yu, 2024) to predict the maximum wave height (or $h_{\max}$) and the wave height time series (or $\eta(t)$) at one station using various ML models. This study demonstrated, among other things, that a CNN model using only the initial surface elevation snapshot was able to predict $h_{\max}$ for one station with high R^2 .

Our primary aim here is to extend the $h_{\max}$ predictions to as many stations and scenarios in our dataset as possible and to add first wave arrival time predictions. We choose to focus on these targets because they already provide highly valuable information and are enough for early warning systems (Rodríguez et al., 2022). We use the same input mentioned above in addition to the bathymetry, the station coordinates and the scenario source earthquake slip distribution type (or slip type), our surrogate is therefore a surrogate for the Boussinesq solver. We do not replace the Okada model since it is inexpensive compared to the propagation and only makes learning more difficult. Additionally, we add aleatoric uncertainty quantification via the probabilistic heads and assess the quality of these estimates using metrics which we define in the Evaluation Metrics section below.

Study Area

The Makran subduction zone lies in the Arabian sea along the coasts of both Pakistan and Iran. The region is tsunami prone but has been understudied because of its low population density (Salah et al., 2021). It remains however important from an economic and strategic point of view, both the Iranian and Pakistani governments have been pursuing multiple major development projects on the coastal areas since 2016, and these have also seen a steady influx of migrants during this same period. All of this makes the wide availability of tsunami hazard assessments and early warning systems more and more crucial.

Figure 1 below is a bathymetric map of the area under study. It is of somewhat low resolution and is the same as the one used in Yu (2024). A structured mesh divides the whole domain into 160 segments which are used in the generation of different scenarios.

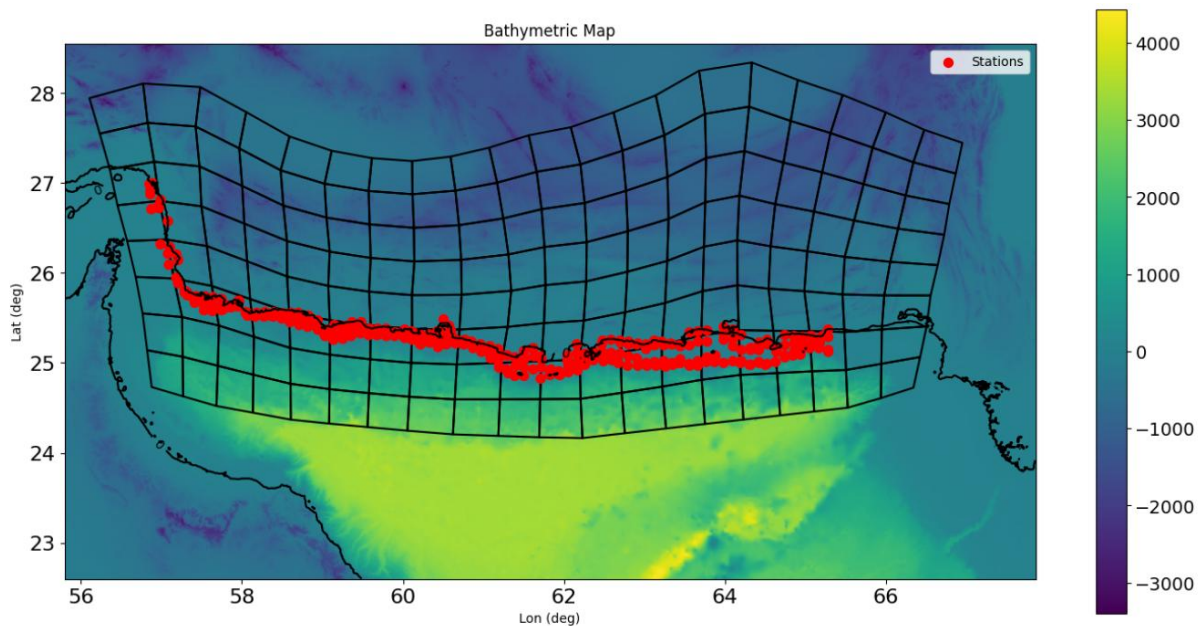


Figure 1: Bathymetric map of the Makran Subduction Zone with station locations.

As can be seen, we have a large number of stations (322 in total) placed along the entire coast where the outputs of the numerical simulations, detailed in the section below, are recorded.

Dataset

Overview

Our dataset comprises 2922 scenarios which correspond to different source earthquake parameters, and includes for each a unique initial surface elevation snapshot (generated from the parameters using an Okada model and a Kajiura filter and of size 992×2010 just like the bathymetry), in addition to a wave height ($\eta(t)$) and two velocity components (u and v) time series for each of the 322 stations along the coast.

The time series come from a Boussinesq simulation (with 8987 time steps spanning approximately 4 hours) and from them we extract the maximum wave height and the first wave arrival time for every station. We define the arrival time as the earliest time at which $|\eta - \eta_0| \geq 0.5$ cm with η_0 representing the initial water level and η representing any subsequent water level as done in a similar study (Rodríguez et al., 2022). Additionally, we only use 317 out of the 322 stations as we skip those where $\eta(t)$ is constant across all scenarios.

There is a wide range of source earthquake magnitudes in our dataset and Table 1 shows the distribution of scenarios per magnitude.

Magnitude	Number of Scenarios
7.9	126
8.0	123
8.1	118
8.2	119
8.3	111
8.4	96
8.5	89
8.6	84
8.7	77
8.8	50
8.9	644
9.0	668
9.1	418
9.2	199

Table 1: Number of tsunami scenarios per earthquake magnitude.

As can be seen, there are significantly more scenarios coming from large magnitude earthquakes. The scenarios can also be classified according to the slip distribution of the source earthquake, homogeneous or heterogeneous, with all heterogeneous scenarios

being of the higher magnitudes (8.9 to 9.2) and all homogeneous ones of the lower magnitudes (7.9 to 8.9). Magnitude 8.9 is the only one where we have both slip types, with 50 homogeneous and 594 heterogeneous scenarios. The addition of these 1879 heterogeneous scenarios is where our dataset deviates from the one seen in Yu (2024).

Table 2 shows some target related statistics to give an idea of their scale. We can also see $h_{\max}$ is generally larger and shows greater variability in heterogeneous slip scenarios (and larger magnitudes) whereas arrival time is generally smaller but also shows greater variability for these same scenarios. We only show a few representative magnitudes here but a more detailed breakdown can be found in Appendix 1.

Category	$h_{\max}$ median (m)	$h_{\max}$ IQR	arrival time median (s)	arrival time IQR
All	0.872	1.616	556.1	745.2
Homogeneous	0.515	1.164	782.0	410.2
Heterogeneous	1.102	1.754	290.0	705.0
Mw 7.9	0.405	1.122	814.0	254.0
Mw 8.6	0.590	1.145	753.2	642.6
Mw 9.2	1.506	1.961	157.0	453.5

Table 2: Median and IQR for $h_{\max}$ and arrival time per slip type and per magnitude (in metres and seconds).

Preprocessing and Split

This is a brief subsection outlining the preprocessing methods that were applied to our dataset and describing how the train/test split was made.

We applied z-scoring to the fixed bathymetry grid, robust scaling (median/IQR) to the initial surface elevation snapshots and transformed the targets using $\log_{1p}$ for arrival time and arcsinh for $h_{\max}$ before applying z-scoring as well. These choices were informed by training data distributions and all scaling was done using training set statistics only. The station xy coordinates are normalised to $[-1, 1]$ along each axis and augmented with sinusoidal Fourier features before being fed to the model.

The train/test split was done at the scenario level using magnitude stratification to deal with the issue of class imbalance briefly mentioned in the previous subsection. This was necessary because we are also assessing performance by magnitude, and with a random train/test split there is always a chance that at least one magnitude class will be severely underrepresented. Magnitude stratification can also make it more likely for extreme events to appear in all of the sets, as there usually is a correlation between the severity of a tsunami event and the magnitude of the source earthquake (as also seen in our dataset).

Some other stratification methods were also considered in addition, such as applying PCA to the initial surface elevation snapshots and clustering the PCA scores using k-means, but these were ultimately abandoned in favour of magnitude stratification alone as it is simple to implement correctly and easy to interpret.

We also decided to mostly confine ourselves to the use of two validation folds rather than doing the more extensive 5-fold cross validation initially planned to reduce computational costs. For the final run, we had 85% of the scenarios in the development set, with 7.5% of those used as micro validation, and the remaining 15 % in the test set.

Model Architecture

We decided to go for a dual task model rather than two separate models for $h_{\max}$ and arrival time as in Rodriguez et al. (2022) despite it being potentially more difficult to tune because this proved beneficial in some early tests. This can be explained by the fact the two tasks are related.

Following Yu (2024), the first dual task model we tried was a CNN backbone to which we added slip aware probabilistic heads (via embeddings), and it only required the initial surface elevation snapshot, the bathymetry and the scenario slip type as inputs. Its performance was not satisfactory, the mean predictions were off and it was poorly calibrated which only served to confirm the complexity of the task. We then moved to a model using a ResNet style CNN (ResNet18) as backbone and taking the same inputs, and saw an improvement in the results which is why we use a modified version of this as our encoder in the final model. ResNet style CNNs have been used in various geophysical contexts such as satellite flood detection (Rambour et al., 2020) and ocean wave and wind retrieval from SAR images (Xue et al., 2023). Their growing popularity comes from the fact they successfully deal with the two issues of vanishing gradients and degradation (train error going up with the addition of layers) which are often encountered in multilayered models including plain CNNs (He et al., 2016). This is achieved through the use of skip connections and allows for deeper models and the extraction of richer features which is of great benefit for our purpose of multi station prediction.

ResNet18 includes 17 convolutional layers and a fully connected one (which we omit in our model), to which are added Batch Normalization, ReLU activation and pooling layers (two including one before the final fully connected layer). These are mostly organised in multiple blocks which learn a residual mapping, corresponding to the difference between the ideal mapping and the identity one, and output the sum of the predicted residual and the received input (projected or passed unchanged). The addition of the input is how the skip connections we mentioned are implemented.

From the ResNet18 encoder we get a global feature vector $g \in \mathbb{R}^{512}$ which is the output of the Global Average Pooling layer, in addition to a spatial feature map $F \in \mathbb{R}^{512 \times 31 \times 63}$ which is the output of the layer just preceding it (and of the last residual block of the model). We concatenate slip type embeddings (controlled by learned scalar gates), station ID embeddings and Fourier and normalised station coordinate features with g and sample F to build local context features for every station.

The sampling is done by creating average pooled versions of F using kernel sizes of 1 (no pooling), 3 and 5 and then bilinearly sampling the three versions at the station locations to extract three vectors of size 512 for every station.

These are also concatenated with the inputs previously mentioned and we feed the resulting per station vectors into a shared station MLP decoder with probabilistic heads. We use Student-t distributions with learned degrees of freedom for each task and slip type (4 in total). The outputs of the MLP are then calibrated using learned per station and per slip

biases for the means, and tanh bounded per station and per slip adjustments for the log scale before it is exponentiated. The resulting mean (μ) and scale (σ) outputs are in the standardised (or z-scored) transformed space.

Last thing to note about the model is that we apply dropout on both g and station ID embeddings before concatenation. The former was done to improve performance and prevent overfitting whereas the latter was also done out of a desire to avoid redundancies with learned station biases.

We also considered using Okada parameters as inputs but these were dropped as they did not improve performance.

Evaluation Metrics

This section serves as a brief introduction to the evaluation metrics we will be using. We are assessing the median our model predicts as we would point predictions using MAE but we are also assessing the predicted (or predictive) distribution itself and the quality of our model's calibration with the following metrics:

- **PICP (Prediction Interval Coverage Probability):** The percentage of targets that fall within their respective prediction intervals. For a 95% prediction interval, PICP should be close to 95% or else our prediction intervals are not reliable.
- **NLL (Negative Log Likelihood):** Likelihood corresponds to the probability (predictive density here) of observing the data we have under the model's predictive distribution. The higher Likelihood is, the better our model. This can be turned into a loss function to be minimised by taking the log (for numerical stability) and the negative. And this is exactly what we use as training loss here.
- **CRPS (Continuous Ranked Probability Score):** A generalisation of MAE to probabilistic forecasts, it measures the distance (integrated squared difference) between the entire predictive CDF and the target or more specifically a step CDF which takes the value of 1 starting from the target (also known as Heaviside step function). It is more robust than NLL to outliers and also rewards both sharpness and good calibration.
- **PIT (Probability Integral Transform):** The percentile of a target value calculated using its own predictive cumulative CDF (built from the predictive distribution). If the model is perfectly calibrated, the distribution of PIT values will be the uniform distribution on $[0, 1]$.
- **Normal Q-Q Plot (quantile-quantile plot):** A plot that compares the quantile (Dunn-Smyth) residuals of our dataset to the theoretical normal quantiles. The quantile residuals are calculated by taking the PITs of all targets in our dataset and transforming them using the inverse CDF of a standard normal distribution. If the quantile residuals do not closely follow the 45 degree line then the model is miscalibrated or misspecified.

- **IW95 (Interval Width):** The average width of the 95% prediction interval. It can give an idea of how sharp the model's predictions are.
- **CBP (correct band probability):** The probability our model assigns on average to the physical band where the $h_{\max}$ target actually lies. Here we define 9 bands separated by the following boundaries: 0, 0.5, 1, 2, 3, 5, 7, 10. Values greater than 10 and those lower than 0 also have their own bands. The choice to have the last boundary at 10 was made to ensure at least 200 values in each band.
- **TWP (time window probability coverage):** The probability our model assigns on average to a window around the true arrival time value. Here we chose 2 min (on either side), 5 min and 10 min windows.

Results

Predictive Performance and Decision Metrics

Here we present MAE, CRPS (in standardised transformed space) and NLL (in physical space) for completeness, but focus on the former two to assess predictive performance for both tasks overall and by slip type. We also look at CBP and TWP since they are actionable metrics.

Slip type	MAE_h (m)	NLL_h	CRPS_h	MAE_t (s)	NLL_t	CRPS_t
All	0.242	-0.103	0.13	133.0	5.83	0.12
Homogeneous	0.196	-0.350	0.12	139.9	6.20	0.10
Heterogeneous	0.268	0.040	0.14	129.0	5.62	0.14

Table 3: Predictive performance metrics (with h for $h_{\max}$ and t for arrival time).

As we can see in Table 3, MAE is higher for heterogeneous scenarios in the case of $h_{\max}$ but slightly lower in the case of arrival time. This is expected since tsunamis generated by earthquakes with higher magnitudes can be more extreme, the waves can be larger and reach the coast earlier. The results are reasonably good especially compared to the relevant IQRs but our MAE for homogeneous scenarios for example is still higher than the one found in Yu (2024) for the same scenarios (0.075 vs 0.196). Adding 316 stations with different local conditions does make the task more difficult.

The CRPS values are however quite low for both tasks indicating that the predicted distributions successfully capture useful information. The learned degrees of freedom (ν) for our Student-t distributions were around 9 and 5.7 for $h_{\max}$ (homogeneous and heterogeneous) and 4.6 and 4.1 for arrival time. We can compare our overall scores (0.13/0.12) to the scores a naïve Student-t reference with the same degrees of freedom per slip as our trained model, zero mean and unit variance (so $\sigma = \sqrt{(\nu - 2)/\nu}$) achieves on our test set: 0.57 for $h_{\max}$ and 0.59 for arrival times. Our scores are ≈ 4 -5x lower which means the forecasts of our model are far better than those of the naïve heavy tailed reference.

Figure 2 shows the performance per station, we report MAE/IQR here to adjust for the scale of the targets and better reflect actual performance (although note the MAE plot looked almost identical). The performance for arrival time is mostly good across the coast and only slightly weaker on the westernmost and easternmost parts. For $h_{\max}$ however we see weaker performance across the entire western half. One possible explanation could be the proximity to the Gulf of Oman coast which can cause a reflection of the waves and make propagation behaviour more complex. Station biases and local context features were added to deal with this issue but it could not be completely resolved.

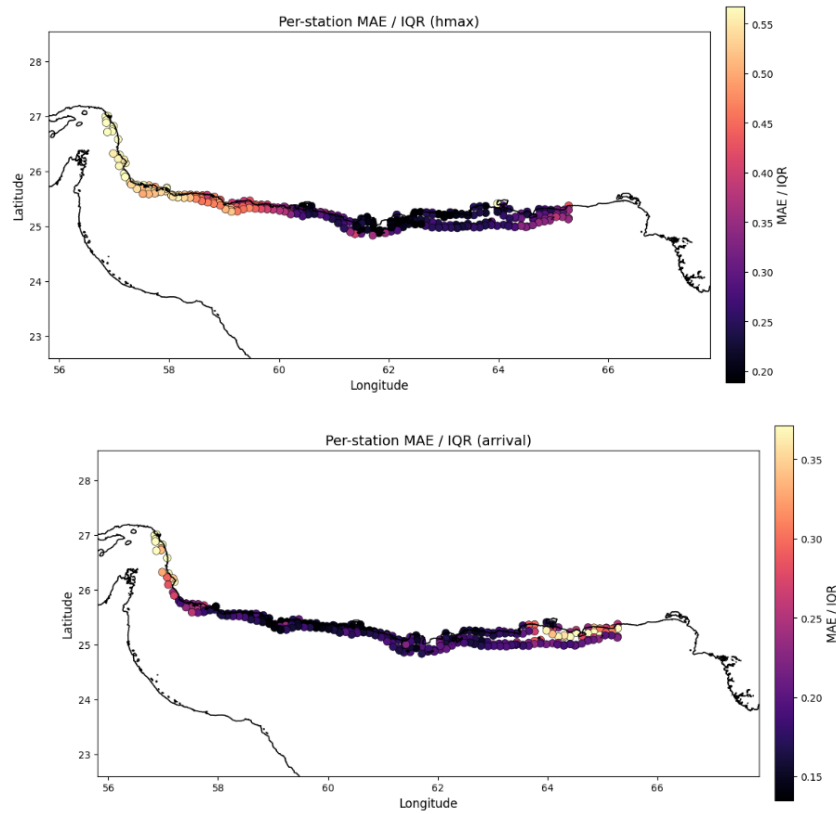


Figure 2: MAE/IQR plot by station for $h_{\max}$ and arrival time.

For CBP and TWP we have the results in Table 4:

Slip type	CBP	TWP (2 min)	TWP (5 min)	TWP (10 min)
All	67%	60%	82%	93%
Homogeneous	71%	51%	80%	94%
Heterogeneous	65%	65%	83%	92%

Table 4: Decision metrics (Correct band probability for $h_{\max}$ and Time window probability for arrival time).

The model assigns a probability of 67% on average to the correct height band for the $h_{\max}$ target (we define the bands as mentioned in the previous section). And it assigns a probability of 82% for values within 5 minutes of the real arrival time on average. These are

reasonably good results and show this model is a solid starting point for an operational early warning system.

Calibration of Distributions

To assess calibration we focus on Q-Q plots, PICP and to a lesser extent PIT from the metrics mentioned earlier.

As we can see from the Q-Q plots in Figure 3, the model is well calibrated, especially for arrival time. The $h_{\max}$ plot shows that the curve follows the 45° line somewhat closely in the middle, however it clearly goes below it on the extreme right and above it on the extreme left. This implies the extreme tails in our model are slightly heavier than they should be. The central slope is also larger than 1 (≈ 1.1) which implies the predicted scale is a bit too narrow. The plot for arrival time also shows the same pattern on the extremes but it is more subtle. The central slope there is also slightly larger than 1 (≈ 1.04). Overall both curves are quite close to the line.

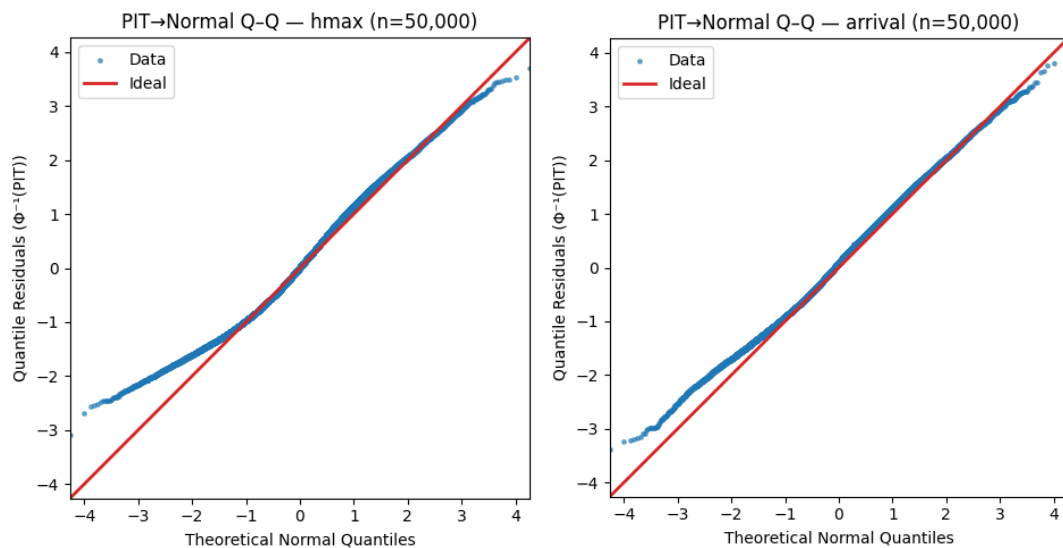


Figure 3: Normal Q-Q plots for $h_{\max}$ and arrival time with both axes standard Normal z-quantiles. Central slopes (within 1 std from the mean) are ≈ 1.1 for $h_{\max}$ and ≈ 1.04 for arrival time.

In Figure 4, the PICP values show good coverage for both tasks overall. There is under-coverage when it comes to the 50% prediction intervals for both tasks and especially $h_{\max}$ (45.5% for $h_{\max}$ and 48.1% for arrival time) but the 95% intervals contain very close to 95% of the values on average and are even slightly conservative (96.6 and 96.2%). This points to the same issue mentioned above concerning the narrowness of the scale and the heaviness of the extreme tails. The PICP patterns per slip are also similar.

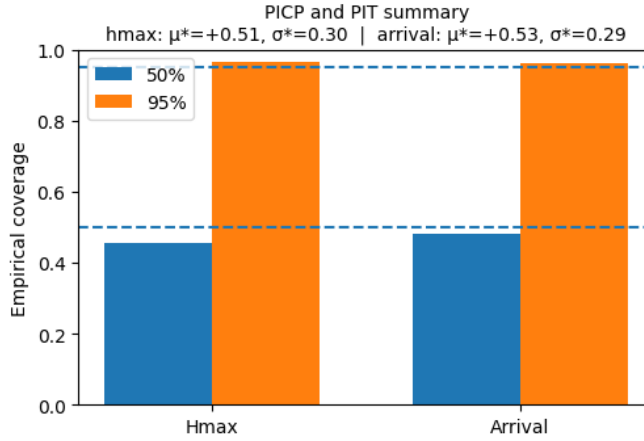


Figure 4: PICP and PIT summary: PICP of 50% and 95% prediction intervals for $h_{\max}$ and arrival time with PIT mean (μ^*) and standard deviation (σ^*) also reported.

The mean (μ^*) and sigma (σ^*) numbers we can also see in Figure 4 come from PIT histograms and are close to 0.5 and 0.289 (0.5 and 0.289 are the μ and σ of a uniform distribution on $[0, 1]$ which is what PIT distributions are in case of perfect calibration). The slightly higher than 0.5 μ^* indicates that the targets tend to fall a bit above the predicted median (in physical space) which points to slight bias, and the slightly higher than 0.289 σ^* points again to the scale and tail issue. PIT histograms by slip can be found in Appendix 2.

Overall, the model is relatively well calibrated which means the predicted distributions are trustworthy and provide valuable information on average.

Uncertainty Quantification

We only focus on aleatoric uncertainty here, which we approximate using our predicted distributions, because of computational and time constraints. Epistemic uncertainty quantification is done via Ensembles or MC Dropout and Ensembles requires the full training of multiple independent models whereas MC Dropout requires the tuning of dropout placements and hyperparameters as well as potentially performing a large number of stochastic forward passes. Given the time and number of runs it took to pick and tune the final model, none of this was feasible.

Table 5 shows the median IW95 per slip and per magnitude (a few representative ones) for both tasks along with the IQR of IW95.

Category	IW95_median_h (m)	IW95_iqr_h	IW95_median_t (s)	IW95_iqr_t
Overall	0.951	1.499	619.0	699.3
Homogeneous	0.618	1.266	708.2	286.3
Heterogeneous	1.123	1.496	346.3	801.9
Mw 7.9	0.597	1.338	727.2	128.7
Mw 8.6	0.635	1.220	696.7	565.9
Mw 9.2	1.484	1.679	201.5	569.5

Table 5: IW95 (median and IQR) overall per task, per slip and magnitude. We use h for $h_{\max}$ and t for arrival time.

We can see the median IW95 is 0.951 m for $h_{\max}$ and 619 s for arrival time overall which means our model is reasonably sharp when it comes to the predictions relative to the IQRs of the targets (overall IQR is 1.616 for $h_{\max}$ and 745.2 for arrival time).

The median IW95 also increases with magnitude for $h_{\max}$ but decreases with magnitude for arrival time and this matches what we saw earlier for MAE (Error was higher for heterogeneous scenarios in the case of $h_{\max}$ and lower in the case of arrival time). IW95 IQR also seems to track target IQR, it is higher for heterogeneous scenarios in both $h_{\max}$ and arrival time (homogeneous/heterogeneous IQR is 1.16/1.75 for $h_{\max}$ and 410/705 for arrival time).

This indicates that the model responds both to scale and to difficulty of prediction and assigns uncertainty in an appropriate way (also note that Spearman correlation between absolute error and IW95 is approximately 0.65 for $h_{\max}$ and 0.75 for arrival time). This can also be seen in Figure 5 which is a plot of median IW95 by station, the stations that are more difficult to predict (based on MAE/IQR plot in Figure 2) have a wider median IW95.

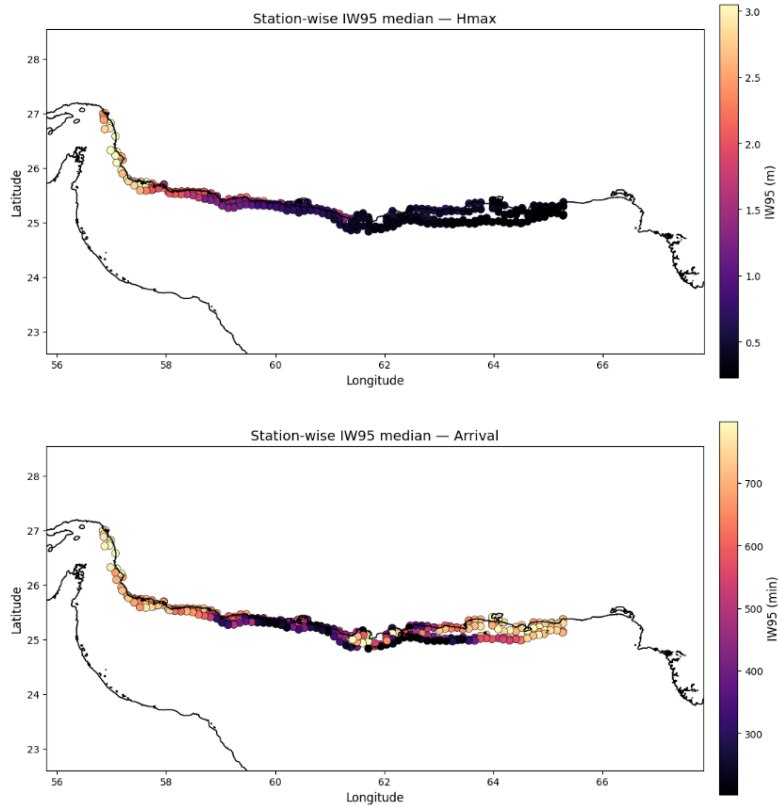


Figure 5: Median IW95 plot by station for $h_{\max}$ and arrival time.

Conclusion and Future Work

In this study we presented a model aimed at serving as an early warning system that predicts both $h_{\max}$ and arrival time, provides full coverage of the coast, deals with diverse tsunami scenarios and quantifies uncertainty.

The model, as already mentioned, is reasonably well calibrated and the probabilistic metrics especially show good performance. This means our uncertainty estimates are useful and reliable. However, our MAE for both tasks remains significantly weaker than what can be seen in Rodríguez et al. (2022) when we take the median as prediction (the errors in Rodríguez et al. (2022) are in the single second and centimetre digits).

This can be explained by a number of factors such as the far greater number of stations we were trying to predict for (317 vs 6 to 12) and their wider range of geographic and bathymetric conditions, the smaller dataset at our disposal (2922 vs 16000) and the much greater diversity of scenarios we aimed to deal with in terms of magnitude and slip type.

The results are nevertheless still encouraging and the paths to improvement are clear. First, the hyperparameter tuning, ablations and comparisons with simpler models explored in development could be extended into a more controlled analysis, with individual changes evaluated separately and the results reported more systematically. Computational and time constraints limited the scope of this analysis. The use of even finer local bathymetric context could also be investigated to address the weaker $h_{\max}$ performance we saw in the western region.

Second, increasing the size of the dataset could be of great help, the great advantage of our dataset was the sheer breadth of its coverage but perhaps some additional granularity is needed for our purpose.

Third, for future operational use, focusing on a smaller set of stations that still spans the coast could be a good idea. This could be achieved by training only on the chosen stations or giving them greater weight, and their selection could be based on spacing, depth and proximity to five of the main coastal cities (Chabahar, Gwadar, Ormara, Pasni and Jask).

Finally, epistemic uncertainty quantification should be added as aleatoric and epistemic uncertainty are both important for a good understanding of the limitations of a model and their sources (Salah et al., 2021). Ensembles and MC dropout are the two possible methods but Ensembles are likely more reliable.

AI Acknowledgement Statement

I used ChatGPT-4o, ChatGPT-o3 and ChatGPT-5 by OpenAI extensively during the literature review, the project planning and the project execution. I asked it questions about papers I read and also to summarize those I didn't, I used it for help with the first versions of most parts of the code and often for debugging. I had many conversations about pretty much every aspect of the project, and I only included here what I can fully own.

Appendix 1

Magnitude	$h_{\max}$ median	$h_{\max}$ IQR	arrival time median	arrival time IQR
7.9	0.405	1.122	814.0	254.0
8.0	0.406	1.082	820.4	253.2
8.1	0.435	1.121	799.6	298.0
8.2	0.474	1.146	828.5	306.1
8.3	0.484	1.134	778.8	365.3
8.4	0.468	1.079	812.5	458.4
8.5	0.528	1.098	748.4	488.7
8.6	0.590	1.145	753.2	642.6
8.7	0.684	1.243	663.4	701.9
8.8	0.771	1.199	552.9	738.0
8.9	0.942	1.575	419.9	745.2
9.0	1.091	1.720	306.1	716.3
9.1	1.197	1.882	216.3	607.4
9.2	1.506	1.961	157.0	453.5

Table A1: median and IQR of $h_{\max}$ and arrival time per magnitude.

Appendix 2

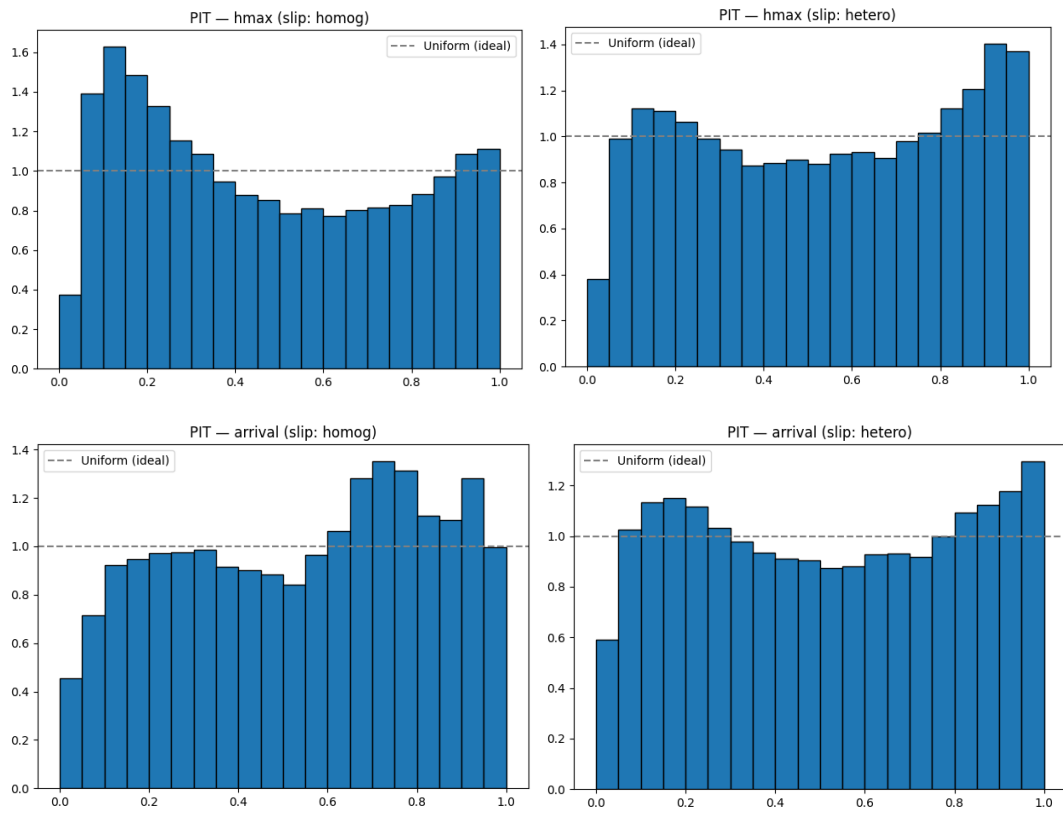


Figure A2: PIT Histograms by task ($h_{\max}$, arrival time) and slip type (homogeneous, heterogeneous).

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